CHEMISTRY

2AB 2013
Booklet 1

Name: ____________________________

Teacher: __________________________

TIME ALLOWED FOR THIS PAPER

Reading time before commencing work:	Ten minutes
Working time for the paper:	Three hours

MATERIALS REQUIRED/RECOMMENDED FOR THIS PAPER

To be provided by the supervisor:

- 2 Booklets
- Multiple-choice Answer Sheet (on booklet 2
- Chemistry Data Sheet

To be provided by the candidate:

- Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, eraser, correction tape/fluid, ruler, highlighters

- Special items: non-programmable calculators approved for use in the WACE examinations

IMPORTANT NOTE TO CANDIDATES

- No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-choice	25	25	45	/50	/25
Section Two: Short answer	11	11	70	/80	/40
Section Three: Extended answer	5	5	65	/70	/35
					/100

Instructions to candidates

1. Answer the questions according to the following instructions.

   Section One: Answer all questions on the separate Multiple-choice Answer Sheet provided. For each questions shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

   Sections Two and Three: Write answers in this Question/Answer Booklet <u>in blue or black pen</u>.

2. When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

5. The Chemistry Data Sheet is **not** handed in with your Question/Answer Booklet.

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Section One: Multiple-choice **25% (50 marks)**

This section has 25 questions. Answer all questions on the separate Multiple-choice Answer Sheet provided. For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 45 minutes.

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1. An element X has the electron configuration 2,8,3. What is the most likely formula of the compound formed when X bonds with sulfur?

(a) XS_3
(b) XS_2
(c) X_2S
(d) X_2S_3

2. Which of the following statements regarding the layout of the periodic table is **false**?

(a) Atomic number increases across the periods
(b) Atomic number increases down the groups
(c) Elements are arranged in order of increasing mass number
(d) Elements are arranged in order of increasing atomic number

3. In which of the following equations is water acting as a Bronsted-Lowry base?

(a) HCO_3^- (aq) + H_2O (l) $\rightarrow$ CO_3^{2-} (aq) + OH^- (aq)
(b) SO_4^{2-} (aq) + H_2O (l) $\rightarrow$ OH^- (aq) + HSO_4^- (aq)
(c) ClO^- (aq) + H_2O (l) $\rightarrow$ $HClO$ (aq) + OH^- (aq)
(d) $H_2PO_4^-$ (aq) + H_2O (l) $\rightarrow$ H_3O^+ (aq) + HPO_4^- (aq)

4. Consider the endothermic reaction below.

$Ba(OH)_2$ (s) + 2 NH_4SCN (s) $\rightarrow$ Ba^{2+} (aq) + 2 SCN^- (aq) + 2 H_2O (l) + 2 NH_3 (g)

Which of the following statements regarding this reaction is **false**?

(a) The products would have a higher enthalpy than the reactants
(b) The heat of reaction would be positive
(c) The energy required to break the bonds would be greater than the energy released when new bonds form
(d) The reverse reaction would have a greater activation energy than the forward reaction

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5. The solubility of sodium chloride is 35.9 g per 100 g of water at 25 °C. Which of the following would form an unsaturated solution of sodium chloride?

(a) 9.04 g of NaCl in 25 g water
(b) 7.01 g of NaCl in 20 g water
(c) 3.60 g of NaCl in 10 g water
(d) 53.85 g of NaCl in 150 g water

6. Most oxygen atoms contain 8 neutrons. What is the most likely electron configuration for an atom of oxygen that has 9 neutrons?

(a) 2,5
(b) 2,6
(c) 2,7
(d) 2,9

7. Compare the gases sulfur dioxide (SO_2) and ammonia (NH_3). Which of the following statements is **true**?

(a) They would have the same molecular mass at STP
(b) They would have the same molar volume at STP
(c) The molecules of each gas would travel at the same speed at STP
(d) 22.71 L of each gas would weigh the same at STP

8. The concentration of magnesium ions (Mg^{2+}) in seawater is approximately 1.292 g L^{-1}. Calculate the number of magnesium ions that would be present in each millilitre of seawater.

(a) 5.306×10^{-2}
(b) 5.306×10^{-5}
(c) 3.20×10^{19}
(d) 3.20×10^{22}

9. Which of the following pairs of compound names and formulas are correct?

(i) Sodium nitrate, Na_3N
(ii) Phosphorus trichloride, PCl_3
(iii) Copper (I) sulfate, $CuSO_4$
(iv) Hydrogen peroxide, H_2O_2
(v) Zinc hydroxide, $Zn(OH)_2$

(a) (i), (ii) and (v)
(b) (i), (ii) and (iv)
(c) (ii) and (v)
(d) (ii), (iv) and (v)

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10. List the species present in a solution of sulfurous acid in order of decreasing concentration (ignoring the presence of water).

(a) H^+, HSO_3^-, SO_3^{2-}
(b) H_2SO_3, H^+, HSO_3^-, SO_3^{2-}
(c) H^+, H_2SO_3, HSO_3^-, SO_3^{2-}
(d) H_2SO_3, HSO_3^-, SO_3^{2-}, H^+

11. Which of the metals below would displace Sn^{2+} ions from solution but not Co^{2+} ions?

(a) Nickel
(b) Copper
(c) Iron
(d) Chromium

12. Which of the following statements regarding catalysts are **true**?

(i) A catalyst increases the rate of a reaction
(ii) A catalyst increases the kinetic energy of the reacting particles
(iii) A catalyst can be recycled and reused
(iv) A catalyst lowers the activation energy of a reaction
(v) A catalyst allows a larger proportion of particles to react

(a) (i), (ii), (iii) and (iv)
(b) (i), (ii), (iv) and (v)
(c) (i), (iii), (iv) and (v)
(d) (i), (ii), (iii), (iv) and (v)

13. Which of the following saturated aqueous solutions would not conduct electricity well?

(a) $Mg(NO_3)_2$ (aq)
(b) $Al(OH)_3$ (aq)
(c) Na_3PO_4 (aq)
(d) K_2SO_4 (aq)

14. What is the correct IUPAC name for the organic molecule shown below?

$$\begin{array}{lllllll} & & & & CH_3 & & \\ & & & & | & & \\ CH_3 - & CH_2 - & CH & - & C & - CH_2 - & CH_3 \\ & & | & & | & & \\ & & CH_3 & & CH_2 - CH_2 - CH_3 & & \end{array}$$

(a) 3,4-dimethyl-4-propylhexane
(b) 3,4-dimethyl-4-ethylheptane
(c) 3-propyl-3,4-dimethylhexane
(d) 4-ethyl-3,4-dimethylheptane

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15. Which of the following substances contains weak intermolecular forces?

(a) Mercury
(b) Diamond
(c) Kerosene
(d) Steel

16. Consider the following three aqueous solutions;

0.2 mol L^{-1} $MgCl_2$ (aq) 0.2 mol L^{-1} KNO_3 (aq) 0.2 mol L^{-1} NaBr (aq)

Which solution would have the highest boiling point?

(a) $MgCl_2$
(b) KNO_3
(c) NaBr
(d) The boiling points would be the same as concentrations are the same

17. A carton of milk will spoil and turn sour much more quickly if it is left on the bench top instead of being kept in the refrigerator. In terms of the kinetic theory, the **best** explanation for this is;

(a) In the refrigerator, the milk is not exposed to such high levels of the bacteria that causes milk to turn sour
(b) In the refrigerator, less milk particles have the required kinetic energy to overcome the activation energy barrier
(c) In the refrigerator, the activation energy for the reaction that makes milk sour is much lower
(d) In the refrigerator, the concentration of bacteria in the milk is lower, creating fewer collisions and therefore a slower reaction rate

18. What is the formula of icosane, a straight chain alkane with 20 carbon atoms?

(a) $C_{20}H_{38}$
(b) $C_{20}H_{42}$
(c) $C_{20}H_{36}$
(d) $C_{20}H_{40}$

19. Which of the following correctly shows the ionic equation (i.e. only reacting species) for the reaction between sodium carbonate solution and hydrochloric acid?

(a) Na_2CO_3 (aq) + 2 H^+ (aq) $\rightarrow$ 2 Na^+ (aq) + CO_2 (g) + H_2O (l)
(b) $NaCO_3$ (aq) + 2 H^+ (aq) $\rightarrow$ Na^+ (aq) + CO_2 (g) + H_2O (aq)
(c) CO_3^{2-} (aq) + 2 H^+ (aq) $\rightarrow$ H_2O (l) + CO_2 (g)
(d) CO_3^{2-} (aq) + 2 Na^+ (aq) + 2 H^+ (aq) $\rightarrow$ 2 Na^+ (aq) + CO_2 (g) + H_2O (l)

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20. In which of the following species does the underlined atom have an oxidation state of +5?

(i) $H\underline{Cl}O_3$
(ii) $\underline{N}O_3^-$
(iii) $H_2\underline{C}O_3$
(iv) $\underline{P}O_4^{3-}$
(v) $\underline{S}O_3^{2-}$

(a) (i) and (iv)
(b) (i) and (ii)
(c) (ii), (iv) and (v)
(d) (i), (ii) and (iv)

21. Which of the following statements regarding covalent molecular substances is **true**?

(a) Covalent molecular substances are formed when electrons are shared
(b) Covalent molecular substances are generally solids at room temperature
(c) Covalent molecular substances will conduct electricity when dissolved in water
(d) Covalent molecular substances have weak intramolecular bonding

22. What products would you expect to form in the catalysed reaction between benzene and bromine?

(a) 1,2-dibromobenzene
(b) 1,1-dibromobenzene
(c) bromobenzene and hydrogen bromide
(d) 1,2-dibromobenzene and hydrogen gas

23. When the leaves of red cabbage are boiled in water, the liquid collected can be used as an indicator. When the red cabbage juice is added to water it turns purple, when it is added to sodium hydroxide it turns green and when it is added to hydrochloric acid it turns red. Which of the following correctly shows the colours you would expect to observe when red cabbage juice is added to the solutions below?

	Lemon juice	**Bicarb soda**	**Vinegar**
(a)	red	green	red
(b)	red	green	purple
(c)	purple	red	purple
(d)	red	red	purple

24. Which of the following is **not** a redox reaction?

(a) $I_2\,(aq) + Mg\,(s) \rightarrow 2\,I^-\,(aq) + Mg^{2+}\,(aq)$
(b) $SO_4^{2-}\,(aq) + 4\,H^+\,(aq) + Zn\,(s) \rightarrow SO_2\,(g) + 2\,H_2O\,(l) + Zn^{2+}\,(aq)$
(c) $Ba^{2+}\,(aq) + SO_4^{2-}\,(aq) \rightarrow BaSO_4\,(s)$
(d) $2\,H_2O_2\,(aq) \rightarrow 2H_2O\,(l) + O_2\,(g)$

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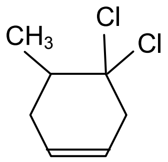
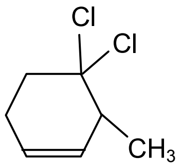
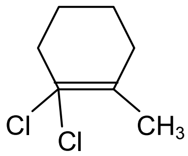
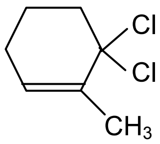
25. Which of the following isomers can correctly be named 3,3-dichloro-2-methylcyclohexene?

(a) 

(b) 

(c) 

(d) 

End of Section One

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